

Lecture 23

Trefoil v4: First Class Functions

Group Work

Add each of the following operators to Trefoil as a **Trefoil function** (if possible) or as a language **primitive** (if needed)

1. Power

- $(\text{pow } 2 \ 3) \Rightarrow 8$
- $(\text{pow } 4 \ 2) \Rightarrow 16$

2. Absolute value

- $(\text{abs } 5) \Rightarrow 5$
- $(\text{abs } -6) \Rightarrow 6$

Group Work

What does each expression evaluate to?

- A. `(lambda (x) (+ x 1))`
- B. `((lambda (x) (+ x 1)) 4)`
- C. `(lambda (x) (lambda (y) (+ x y)))`
- D. `((lambda (x) (lambda (y) (+ x y))) 4)`
- E. `((((lambda (x) (lambda (y) (+ x y))) 4) 5)`
- F. `(lambda (x) ((lambda (y) (+ y 1)) 5))`

Group Work

```
(define y 10)
(define f (lambda (x) (+ x x)))
(define g (lambda (x) (+ x y)))
(define y 20)
(define h (lambda (x) (lambda (y) (+ x y)))))
```

```
(f 10)
(g 10)
(h 10)
((h 10) 10)
```

1. What are y, f, g, and h bound to in the dynamic environment after evaluating these bindings?

2. What does each call evaluate to?